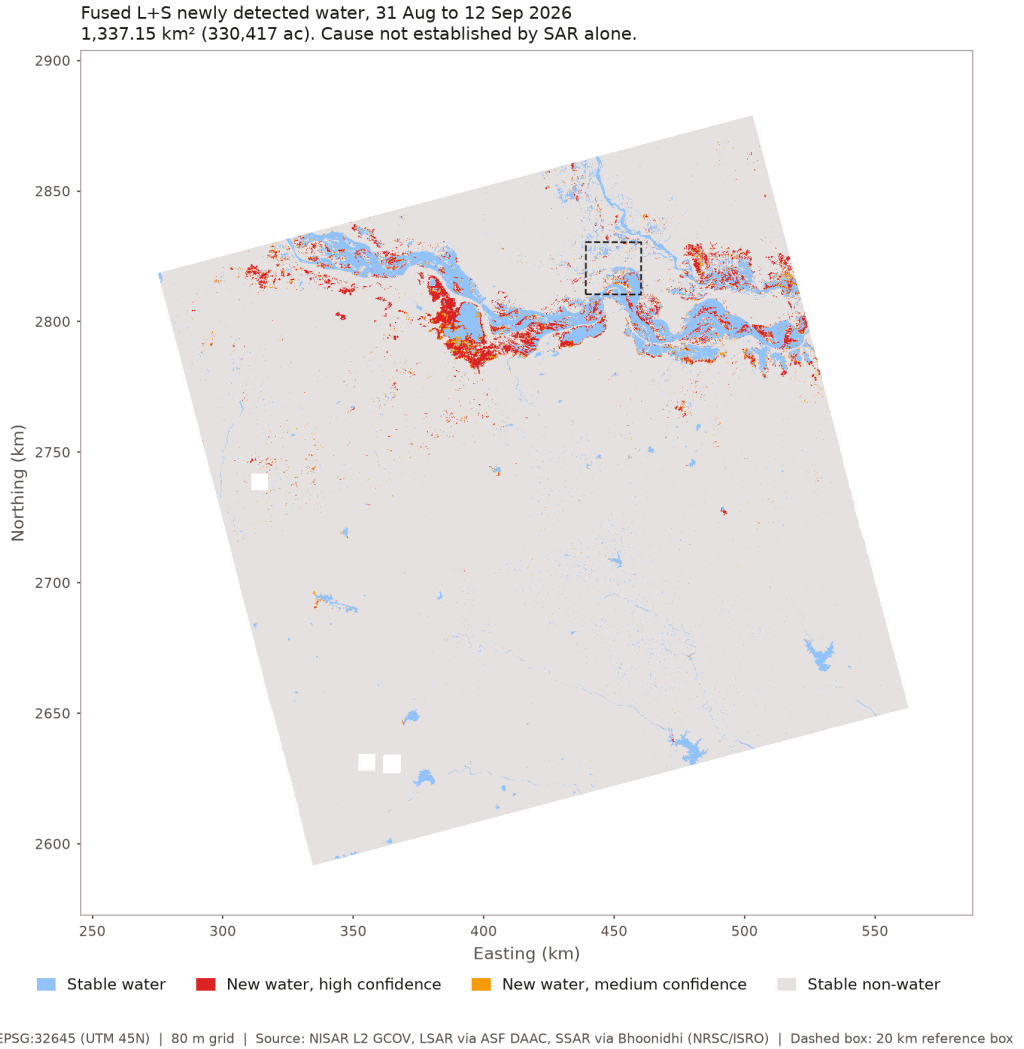
Does combining NISAR L-band and S-band give a better water map than either band alone?

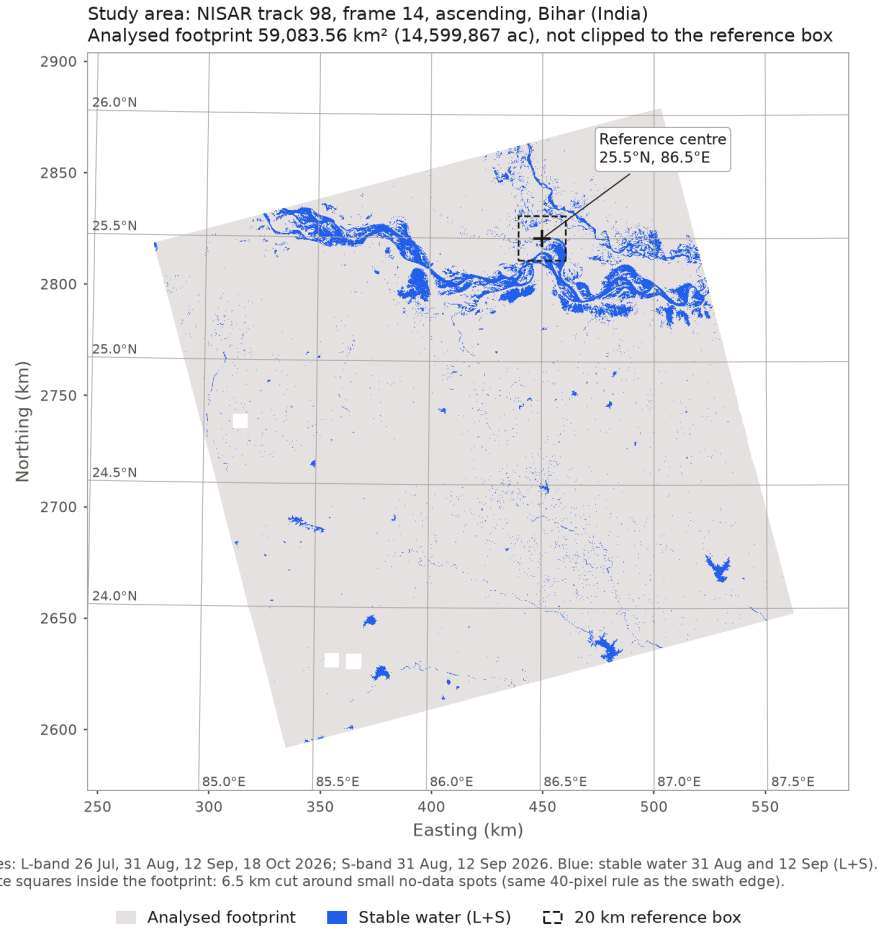
- Full scene, track 98, frame 14: 59,083.56 km² (14,599,867 acres), not the 20 km box
- Three methods on the same 80 m grid and the same dates: L-band only, S-band only, L+S
- Four dates, 26 Jul to 18 Oct 2026, to follow water rising and falling
- Checked against Sentinel-2 optical images taken hours to days from the radar

> **Answer, from the Sentinel-2 scores**
> L+S scored a higher F1 than L-band in all 3 tests.
> L+S did not beat S-band: equal on 12 Sep, lower on 31 Aug and for new water.

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We processed the full NISAR scene; the 20 km box was only for finding it



2026	L-band (1.239 GHz)	S-band (3.2 GHz)
26 Jul	DHDH	not used
31 Aug	DHDH	DHNA
12 Sep	DHDH	DHNA
18 Oct	DHDH	not used

- L-band from ASF DAAC, S-band from Bhoonidhi (NRSC/ISRO); ascending, 10 m grid, EPSG:32645
- Terms used: HHHH for detection, HVHV for fusion and polarization. HHHV (S-band) is not an intensity, so it is not used as one
- L and S were taken seconds apart on 31 Aug and 12 Sep, so the three-way test uses that pair
- Catalogue search: ASF lists L-band full-pol on descending track 120, frame 76 (e.g. 2 and 14 Sep), not processed here. Bhoonidhi showed no S-band compact-pol for the dates searched

One 7-stage pipeline per band; only the thresholds change between L and S

1. **Average to 80 m:** $\bar{P} = (1/n)\ \Sigma\ P_i$ over each 8×8 block, needs n ≥ 32 valid pixels
2. **Check the pair:** same term (HHHH) and product (GCOV) on both dates
3. **Decibels:** $\sigma^0 = 10 \cdot \log_{10}(\bar{P})$; below −40 dB is no data; cut 40 pixels (3.2 km) at every no-data edge
4. **Speckle:** median of the valid 3×3 neighbours
5. **Change:** $\Delta = \mathrm{dB(after)} - \mathrm{dB(before)} = 10 \cdot \log_{10}(P_2/P_1)$
6. **Stable water:** below the water level on both dates. Otsu on Δ is reported; `T_seed = min(−5, Otsu, −3)` stayed −5 dB
7. **New water:** seeds with Δ below the seed level, grown into 8 neighbours below the growth level; remove groups under 8 pixels. Then classes 0, 1, 2, 3, 5

Example: a field going from −8.2 dB to −17.0 dB has Δ = −8.8 dB, so it is a seed (high-confidence new water).

Threshold (dB)	L	S
Stable water, both dates	−13.0	−12.5
Seed change	−5.0	−4.8
Growth change	−2.5	−2.2
Otsu (reported only)	−3.26	−3.47

Set by hand per band: the 24 cm L wave and 9.4 cm S wave see the same surface with different roughness.

> **Memory**
> One scene layer is 1.16 billion values (4.63 GB). The Rust engine reads 1,024-row strips and keeps only a 72 MB grid. Each pair: 44–65 s, 1.87–2.32 GB peak, on a 4-core machine.

Fusion is four if/then rules on one shared 80 m grid

Both grids and their 720 m / 2,880 m offset are exact multiples of 80 m, so L and S pixels line up with no resampling.

1. **Sandbar rule:** L HH < −13 dB but S HH ≥ −10.5 dB on either date → not stable water. A surface can look smooth to the long L wave and rough to the short S wave
2. **Wind rule:** L HH < −15.5 dB and HV < −22.0 dB on both dates → stable water, even if S missed it
3. **Canopy rule:** L HH drops ≥ 3.26 dB, later HV < −20 dB, later HH < −12 dB → new-water candidate. Not verified: optical data cannot see under plants
4. **S-band gate:** every new-water candidate needs later S HH < −10.5 dB

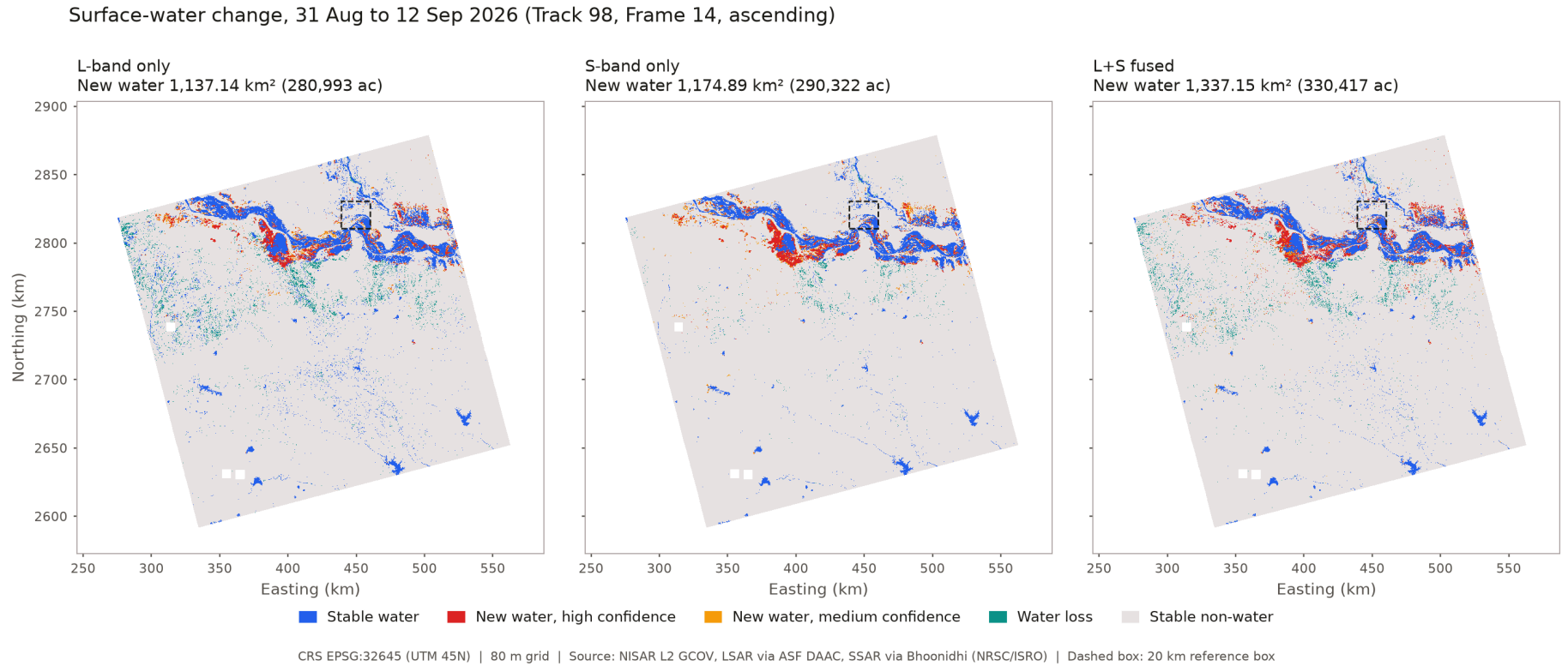
High confidence if both bands found it or L dropped ≥ 5 dB. Rule names describe intent, not proven surface types.

Run	Stable water	New water
All rules	2,652.29 km²	1,337.15 km²
No sandbar rule	2,684.26	1,322.69
No wind rule	2,381.78	1,386.80
No canopy rule	2,652.29	1,113.61
No S-band gate	2,652.29	1,675.85

km². All rules: 655,395 acres stable water and 330,417 acres new water.

- The S-band gate is the strongest control
- Moving it by ±1 dB moves new water between 1,208.44 and 1,449.29 km² (298,611 to 358,128 acres)

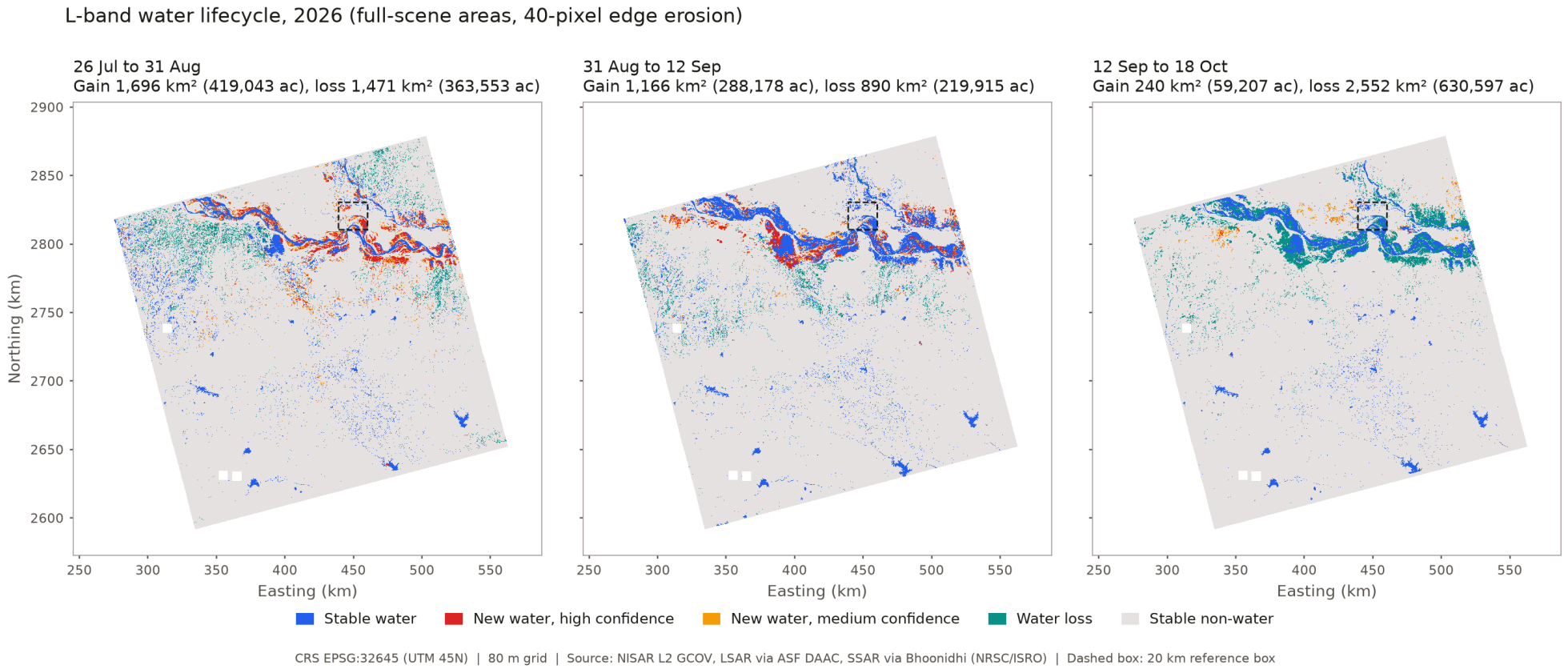
The bands agree on the main pattern; L-band calls far more area water



Same footprint	Stable water	New water	Water loss	Water on 12 Sep
L-band	3,314.83 km² (819,112 ac)	1,137.14 km² (280,993 ac)	888.84 km² (219,637 ac)	4,451.97 km² (1,100,105 ac)
S-band	2,546.37 km² (629,222 ac)	1,174.89 km² (290,322 ac)	214.98 km² (53,123 ac)	3,721.26 km² (919,544 ac)
L+S	2,652.29 km² (655,395 ac)	1,337.15 km² (330,417 ac)	723.37 km² (178,749 ac)	3,989.43 km² (985,811 ac)

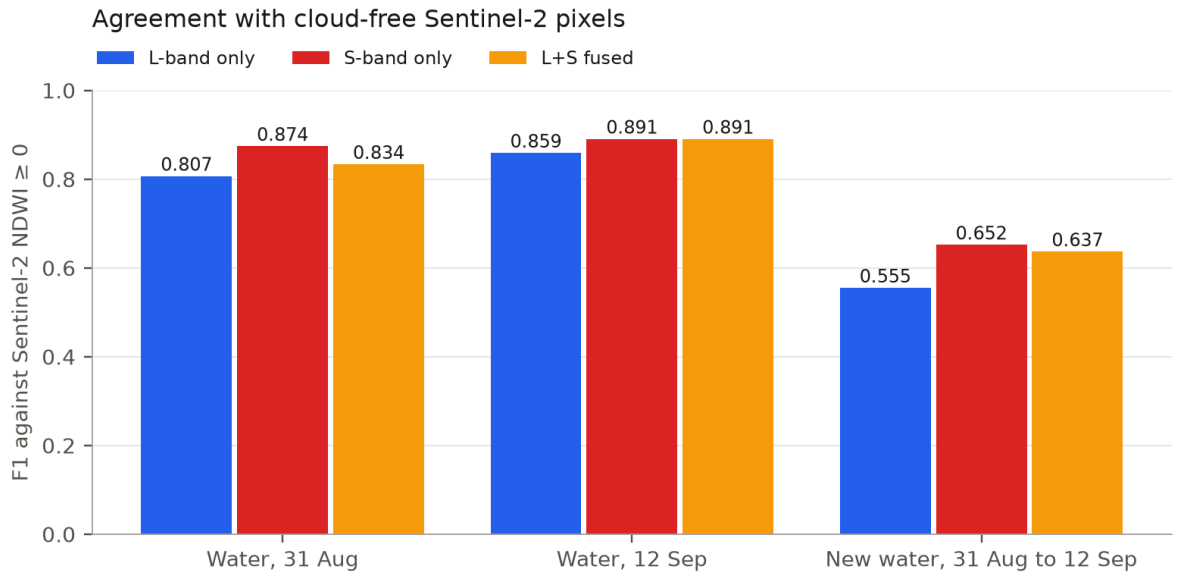
L and S new water overlap by 858.56 km² (212,155 acres), Jaccard 0.59. L+S new water is partly larger by design, because its candidates come from both bands.

Water spread along the river into September; about half the dark water was gone by mid-October



- 26 Jul → 31 Aug: gained 1,695.81 km² (419,043 acres), lost 1,471.24 km² (363,553 acres)
- 12 Sep → 18 Oct: gained 239.60 km² (59,207 acres), lost 2,551.94 km² (630,597 acres)
- L+S new water outside any water seen on 26 Jul or 31 Aug: 1,282.78 km² (316,982 acres), the problem statement's "potential temporary surface inundation"
- SAR alone does not show why water appeared, so no flood or rainfall cause is claimed

L+S beats L-band, mainly through fewer false alarms, but not S-band



Optical labels cannot see water under vegetation. Agreement, not ground truth.

- **Sandbar rule check:** pixels dark in L but bright in S were water only 8.7% (12 Sep) and 9.2% (31 Aug) of the time, against 80.3% and 70.3% where both were dark. Not all sand: bare ground led on 31 Aug, vegetation on 12 Sep
- **Polarization, L-band 12 Sep:** F1 0.848 with HH, 0.835 adding HV, 0.854 with HH + 2·HV

12 Sep (S2 ~5 h later)	Precision	Recall	F1
L-band	0.826	0.895	0.859
S-band	0.911	0.872	0.891
L+S	0.900	0.881	0.891

Scored 7,946.03 km² (1,963,507 acres), about 14% of the footprint, because of cloud.

F1	L	S	L+S
Water, 31 Aug (S2 29 Aug)	0.807	0.874	0.834
New water, 31 Aug → 12 Sep	0.555	0.652	0.637
Water, 18 Oct (L only)	0.744	–	–

Agreement with optical labels, not ground truth. Confusion matrices: VALIDATION.md.

S-band was the strongest single band here; fusion fixed most of L-band's over-mapping

Limits

- Cloud: only about 14% of the footprint could be scored on 12 Sep
- Optical images cannot see water under crops, so L-band's possible advantage there is untested
- Thresholds are hand-set; the S-band gate is the main sensitivity
- New water is a backscatter drop; features narrower than about 240 m are removed
- L-band full-pol (track 120) was found but not processed
- No cause is assigned to any water change

Conclusions

- L+S improved on L-band in every Sentinel-2 test and gave the best new-water precision (0.734)
- L+S did not beat S-band alone in these tests
- The whole scene runs on a normal workstation in about 2 GB of memory
- Next: labels under vegetation, calibrated thresholds, more L+S dates, full-pol data

> **Check it yourself**
> `python3 run_bihar_analysis.py` checks all file hashes, prints every number in these slides and ends with `ALL CHECKS PASSED`.